Topology from the Ground Up

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1 Opening

1.1 What is Topology?

- Show familiar geometry, usually involving lengths, angles, areas/volumes. But some shapes are inherently 'the same'. If we forget about length/angles, circles and squares are pretty much the same shape, just round the corners of the square. We've changed some of the measurable properties of the square (like its angles and area) but haven't changed the fundamental features of the shape, we've just deformed it. Some properties depend on measurement, while others do not. **Topology** is the study of these properties: the features of spaces that remain unchanged under continuous deformation. (Give the example of a torus and a sphere).

- Some transformations are 'nicer' than others. Stretching and bending shapes is nice, they deform shapes continuously. Other transformations like tearing and glueing aren't so nice, they're not continuous. We'd like to investigate these continuous functions, as well as shapes which are actually the same shape under these types of transformations; those that transform shapes without changing their fundamental features.

- Before we start talking about transformations mathematically, we just need to establish a language for talking about collections of objects, and rules that map one object into another.

1.2 Some Preliminaries

- Everything in maths is described using **set theory**.

- A **set** is just a collection of elements. We write a set as a list of its elements contained within curly brackets $S = \{a, b, c\}$. Sets can contain anything (genuinely, anything) like numbers, or points in space, functions, or every cat in the UK (maybe put some stupid joke here idk). Generally, we will use capital letters and fancy letters to denote sets, and lower case letters as elements within those sets (though there are lots of exceptions).

- If we want to describe an element inside of a set, we write $a \in A$ to mean the set A contains the element a.

- If we want to talk about a set of elements that's contained within a another set, this is called a subset. For instance, the set $\{a, b\}$ is a subset of $\{a, b, c\}$. We denote subsets using the $\subset$ / $\subseteq$ symbols, which work very similarly to the less than (or equal to) symbols: $\{a, b\} \subset \{a, b, c\}$

- Because this is a maths video, we'll be looking at lots of sets of numbers. Some commonly used sets which we'll see a lot of

  - $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$ the set of natural numbers
  - $\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$ the set of all positive and negative integers
  - $\mathbb{Q} = \{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\}$ the set off all fractions of integers
  - $\mathbb{R}$ the set of all real numbers (every number you'd find on the number line, the actual definition is very long)

- We will be looking functions a lot and ways of mapping elements in one set to another set, so it's important to grasp how function notation works. For instance, the function denoted f that squares real numbers can be written as $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$. Here the first $\mathbb{R}$ is called the domain of the function (aka the input), and the second $\mathbb{R}$ is the codomain (or range) of the function (aka the output). f takes an input value x from the domain, and will output something that's an element of the codomain.

- I think that's everything but will probably need to add more.

1.3 Less Yap about Preliminaries

- How do we describe shapes mathematically? We use an extremely useful tool in maths called **sets**.

- A set is just a collection of things - we call elements. We can write a set as a list of its elements inside of curly brackets, and maybe give it a label as well $S = \{a, b, c\}$.

- Sets can contain literally anything: numbers, colours, Rubik's cubes, dogs, etc... Since this is a maths video, we're mainly concerned with sets containing numbers. In general, we typically label sets using capital letters, and elements within a set as lower case letters. One set we will use extensively is the set of real numbers $\mathbb{R}$; think of $\mathbb{R}$ as the set of all numbers on the number line.

- To describe maps/functions/transformations (I use these words interchangeably), we first give the function a label, for instance simply f for function, and describe which set f takes as an input, and what it should spit out as an output. For instance, we can have $f : \mathbb{R} \to \mathbb{R}$, which describes a function which takes a number as input, and outputs another number. In order to describe the transformation, we use function notation. $f(x) = x^2$ means our function f will take an input number x, where x is just any number, and will spit out the square of that number, which is also a number.

2 Continuity

2.1 Real Functions

- We want to study 'nice' functions/transformations (I'll use the two words interchangeably), that is transformations that are **continuous**. Let's start by looking at simple one-dimensional functions, like our x^2 example, before we start looking at more complicated functions. Given the example $f(x) = x^2$, think of this as a transformation of the real number line. This is nice, everything is well-behaved, we could draw the resulting graph as a single continuous curve. Now take the Heaviside step function $h(x)$. This is sort of like tearing the real line into two, and we can't draw it as one continuous line. Transformations that 'jump' like this are ones we'd like to avoid, so how can we mathematically filter out these types of transformations? What makes functions 'continuous'?

- Intuitively, and my first idea as well, is that a continuous function should take every value between two points. By that I mean given two points on the graph, a continuous function should take every value between their heights somewhere between those two points. More formally, if c lies between $f(a)$ and $f(b)$, then there exists $x \in [a, b]$ such that $f(x) = c$. This means we should be able to draw the function as a single continuous line without any jumps. This is referred to as the Intermediate Value Property (IVP), and seems like a good definition for a function to be continuous. However, does this actually capture the properties we'd like a continuous function to have?

- Take the function
$$f(x) = \begin{cases} \sin(\frac{1}{x}) & x \neq 0 \\ 0 & x = 0. \end{cases}$$
Drawing the function out, we see it's like a sin curve that oscillates more and more as we approach 0. No matter how small an interval we take around 0, the function still reaches every value between -1 and 1, and so satisfies the IVP. However, something is off as we approach 0.

- As we approach 0, the oscillations are more and more frequent; in fact, they never settle down as we get closer and closer to 0. When we zoom into a continuous function, we expect the function to look flat, we expect the function to relax to a single value, but this function never does. It just oscillates faster and faster as it gets nearer and nearer to 0, so at what point if we're drawing the line do we connect it to $f(0) = 0$? The IVP tells us that a continuous function cannot skip values, but we need to also consider if a function behaves well around a single point.

- This tells us something quite important about continuity; it is a local feature and not a global one. What I mean by that is that continuity is a property possess by each individual point on the graph, not by larger regions as stated in the IVP. As we zoom in closer and closer to a point, the function *should* settle towards a single value, that's the key idea that makes the above example feel discontinuous, despite having the intermediate-value behaviour we wanted. So let's rework our definition again to include this locality condition. To do this we introduce the idea of a **limit**.

- To describe this behaviour mathematically, as a value x approaches a particular value a, we'd like the value of $f(x)$ to also approach some particular value l. Notationally we write this as

$$\lim_{x \to a} f(x) = l.$$

What we're saying here is 'I can make $f(x)$ as close to l as I want, provided that x is close enough to a'. We can describe values that are close together by seeing whether their difference is small, so we can say $f(x)$ is close to l if $|f(x) - l|$ is a small value (the absolute value needed so that the difference is always a positive value). So to write mathematically our statement, we're saying given any small positive value ε, there is some other small positive value δ such that if $|x - a| < \delta$, then $|f(x) - L| < \varepsilon$. Notice the direction of the implication: we control the input, and this forces the output to behave. Continuity says small changes in the input cannot create large changes in the output. We want stability of the output under perturbations of the input. If the implication was the other way around, then most bounded functions would be classified as continuous. With this precise implication, limits behave how we want them, just written in formal mathematics now. We're nearly there to describe continuity, there's just one small condition we're missing.

- For a continuous function, we would like l to be equal to $f(a)$ in this limit. So as we approach a, we should approach the value of the graph. However this isn't always the case.

- Consider the function

$$f(x) = \begin{cases} x^2 & x \neq 1 \\ 2 & x = 1. \end{cases}$$

Here we clearly have a discontinuous function; we've removed a single point from the otherwise continuous graph. However, as x approaches 1, in the limit, the value of $f(x)$ also approaches 1, despite $f(1) = 2 \neq 1$. This is the final idea for our definition of continuity, which should obviously be that at the limit point, the value of the limit should be the value of the function itself i.e. $\lim_{x \to a} f(x) = f(a)$. If this is satisfied for all a in the domain of our function f, then we can safely say f is continuous.

- All together, including using our mathematical notation, we say that a function f is continuous at a point a in the input if:

$$\text{for all } \varepsilon > 0, \text{there exists a } \delta > 0 \text{ such that, if } |x - a| < \delta, \text{then } |f(x) - f(a)| < \varepsilon.$$

To reiterate, we're saying that $f(x)$ and $f(a)$ are close together, provided that x and a are also close together. A function is continuous everywhere if this is true for all values within its input set (i.e. for all possible values of a).

2.2 Continuity in More Abstract Spaces

- How can we expand this idea to higher dimensions? So far we've just looked at the continuity of one-dimensional functions - that's functions which take one real number as input and spit out a real number. You can think of these as equivalent to transforming the space that is just the real number line. When I say 'space', I'm referring to the set as a collection of points. So how can we take our current idea of continuity, and extend to higher dimensional spaces?

- We can create a two-dimensional space by placing a copy of $\mathbb{R}$ onto each point within $\mathbb{R}$. This gives us what is known as the 2D coordinate plane, where each point in this space is described by two numbers. We refer to this space as $\mathbb{R}^2$ since it's basically two copies of $\mathbb{R}$. We can keep adding copies of $\mathbb{R}$ to get $\mathbb{R}^3$ and so on...

- If we first focus on standard 2D and 3D coordinate spaces $\mathbb{R}^2$ and $\mathbb{R}^3$ respectively, it's quite easy to tweak our definition of continuity to work in these contexts. In $\mathbb{R}^2$ and $\mathbb{R}^3$ we can find the distance between two points/vectors in these spaces by taking their difference, and then taking the norm of the resulting vector. The norm is just the length of the resulting vector, which can be found by using simple Pythagoras. So now given a function $f : \mathbb{R}^3 \to \mathbb{R}^3$, we can say f is continuous at $\vec{a} \in \mathbb{R}^3$ if

$$\forall \varepsilon > 0, \exists\, \delta > 0 \,\text{s.t. if } \|\vec{x} - \vec{a}\| < \delta, \text{then } \|f(\vec{x}) - f(\vec{a})\| < \varepsilon.$$

- However, some spaces are a bit stranger. For instance, we can consider the London tube map as a space of points - each station - and the distance between each station the time it takes to get between stations. The distance between stations isn't necessarily the norm of the different between the stations any more; this is a new notion of distance.
- In fact, the only thing we really need to define continuity given our previous definition for $\mathbb{R}$ is for a space to have points in it (duh) and a well-defined distance that can be measured between any two points in that space. Spaces are easy to find, literally any set can be considered as a space of points, we just need a way to define the distance between any two points that doesn't contradict our idea of what 'distance' means.
- Let's say for some generic space X, we'd like to define a function on this space, $d : X \times X \to \mathbb{R}$, that takes the input as two points in our space X, and tells us the distance between those points as a real number. For instance, if our space $X = \mathbb{R}$, then defining $d(x, y) = |x - y|$ gives us the distance function we've already seen for the real line! Abstracting like this allows us to still capture the properties of this distance, but now with the benefit of allowing us to replace the real line with something much more general.
- In fact, we can define much more abstract notions of distance now. Consider the space X that contains every word in the English language, and define the metric $d(x, y)$ to be the number of letters required to be added, removed, or changed to get from one word to the other, i.e. $d(\text{cat}, \text{map}) = 2$ since it requires two letter changes. Although now we're not talking about the distance between points living in real space, this idea of distance still makes sense within this context. What we want to examine is what makes these two examples of distances 'distances'? What properties do they share to guarantee our notion of distance is still preserved even for crazy abstract spaces.
- This is where we establish some **axioms**. An axiom is simply a rule or property that is always true when talking about something. In our case, we would like to establish the rules for which a distance function makes sense to measure distances. Axioms are rules that we - as humans - come up with; they're properties that are naturally desirable, and we usually want the simplest set of rules that gives us what we desire.
  - First off, the distance between two points should always be positive; how would it make sense to have two points which have a negative distance apart? Moreover, the distance between two points should be zero if and only if those points are actually the same point. If we're trying to measure the distance from a point to itself, there isn't any distance to travel to get there; we don't need to move anywhere to stay where we are. To compile these two points mathematically, we are saying that for any two points $x, y \in X$,

    $$d(x, y) \geq 0, \text{ with equality if and only if } x = y.$$

    We use the phrase 'if and only if' to mean the implication is true in both directions: if $x = y$ then $d(x, y) = 0$ AND if $d(x, y) = 0$ then $x = y$. This is necessary because let's say our space $X = \mathbb{R}^2$ and we'll define the distance function as $d((x_1, y_1), (x_2, y_2)) = |x_1 - x_2|$ only measure the distance in the x-direction. Then $d((0, 0), (0, 5)) = 0$. This axiom prevents our distance function from effectively glueing together two distinct points when we don't want it to.
  - Next, it shouldn't matter which point we 'measure' from i.e. the distance from London to Rome is the same as the distance from Rome to London, fairly obviously. Mathematically we just say for any two points $x, y \in X$,

    $$d(x, y) = d(y, x),$$

    we can swap around the inputs for the distance function. (In some contexts it can actually make sense to remove this axiom if maybe the distance function actually measured time, but for our purposes this is not the case).
  - Lastly we need something that you probably intuitively know should be true, but we haven't explicitly stated it yet: the value that the distance function spits out should be in some sense the **shortest** distance between those two points. But remember, because we're dealing with generic spaces, you can't just say 'the shortest distance between two points is just the straight line connecting them' because generic spaces don't necessarily have straight lines you can just draw on them. (This is probably where an example might fit in nicely). So we need to think harder: what does it mean for the distance between two points to be the *shortest*? Think about it.

– Imagine we're walking from point x to point y, surely if we deviate from the supposed 'shortest' path between these points, then our journey must take - at a minimum - the same **or longer** amount of time to traverse. What are we saying here? If while traversing between London and Rome, we'd like to visit another city say Paris, then the distance from London to Paris plus the distance from Paris to Rome must be greater than or equal to the distance from London to Rome. Why isn't it always longer? Because the intermediate point could be anywhere in our space, it could actually lie on the shortest path connecting x and y, making the deviation actually the same length as not deviating at all. Ok so all together what we are saying is: given any three points $x, y, z \in X$, then

$$d(x, y) \leq d(x, z) + d(z, y)$$

. This is known as the triangle inequality. This is then enough to have distances defined in a way that actually makes sense for generic spaces!

- If we pair up a space X with a distance function $d : X \times X \to \mathbb{R}$, notationally written as (X, d) (this isn't the Cartesian product of X and d, just the notation that we have a space X that uses a distance function called d), we call this a **metric space**.

- These three properties are sufficient assumptions to capture our intuition of distance. If we add more requirements, we exclude spaces that still behave perfectly well. If we remove any of them, our notion of distance starts behaving strangely (Try for yourself!). As mathematicians, we want to capture the weakest conditions that still capture behaviour we care about.

- To see how metrics actually differ from your probably pre-conceived notion of distances in Euclidean spaces, let's define a metric space with X being any set, and metric

$$d(x, y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases}$$

This is called the discrete metric. This says that every single point is distance 0 from itself, and every distinct pair of points are exactly a distance 1 apart. Check for yourself that the metric satisfies all of the metric axioms. This example shows how we've stripped apart distance from geometry now; not all metrics will have a nice geometrical interpretation, yet distances within them still make sense. These axioms are the bare minimum of what makes distances make sense.

- Now we have a way of defining more abstract spaces that have a well-defined distance function, we can just use our already established definition of continuity again to define continuous transformations of metric spaces, by replacing the norms with the distance function now! So given two metric spaces (X, d_X) and (Y, d_Y) (the two spaces' metrics aren't necessarily the same), we say a function $f : X \to Y$ is continuous at $a \in X$ if

$$\forall \varepsilon > 0, \exists\, \delta > 0 \,\text{s.t. if}\, d_X(x, a) < \delta, \text{then}\, d_Y(f(x), f(a)) < \varepsilon.$$

2.3 Do we need distance?

- We want to analyse continuous functions even more closely. Notice that in our ever changing definition of continuity, we've always had to establish and way of measuring distance in whatever space we're working in (whether that's norm or metric spaces). What if we could somehow strip away the use of distance in the definition? Like in the example of the square and the circle, clearly these shapes have VERY different properties involving lengths and angles, yet (as said before) fundamentally they are the same shape, the circle being a slightly squished square, or the square a pointy circle.

- Let's look back at the definition we had for continuity for metric spaces: $f : X \to Y$ is continuous at $a \in X$ if

$$\forall \varepsilon > 0, \exists\, \delta > 0 \,\text{s.t. if}\, d_X(x, a) < \delta, \text{then}\, d_Y(f(x), f(a)) < \varepsilon.$$

Take a look at the statement $d_X(x, a) < \delta$. What is this saying? It's asking us to consider all of the points x that are within a distance δ of a. All of the points that are equidistant from some origin point is precisely the definition of a ball, so this says to consider all of the points contained within the ball of radius δ centred at a. Specifically in this case, they're

contained within the **open** ball centred at a, the term open meaning that we don't consider the outer 'skin' layer of the ball, just everything inside of the outermost layer (If instead we had a $\leq$ sign, this would dictate a **closed** ball, which does contain that outer skin layer as well). Likewise $d_Y(f(x), f(a)) < \varepsilon$ tells us to consider all points $f(x)$ that sit inside the open ball of radius ε centred at $f(a)$.

- In fact, in our definition we're not really concerned with the actual values of ε and δ, they're just a way to signify we're concerned about the closeness of the points to a or $f(a)$. What we really care about is the collection of points close to a and $f(a)$ - that's all of the points inside of the open balls. So let's get rid of distance now, and just consider the set of points that are 'close' to a and $f(a)$.

- Rewording the continuity definition, what it's saying is for a continuous function, if we take a collection of points around $f(a)$, then there is some collection of points around a that all get mapped inside of that first collection. We've stripped away any sense of distance now, and all we're left with are collections of 'nearby points'.

- Now in the same sense that we could sort of 'pick' which distance functions we'd like to consider in our metric spaces, we can 'pick' which sets we say contain points that are 'close' together. These sets we will call the **open sets** of the space. In the example where metric space was simply $\mathbb{R}^2$ with $d_{\mathbb{R}^2}(x, y) = \|x - y\|$, the open sets are the (unions of) open balls we can draw on the plane, but in general these open sets can be any amount of subsets of the ambient space we're working in X.

2.4 Topological Spaces

- So to generalise, let's create a set, call it $\mathcal{T}$, which contains all of the collections of points we consider to be 'close' together, these will be subsets of our ambient space X. So $\mathcal{T}$ is a set of subsets of X, which we'll call the **topology** on X. Before we get ahead of ourselves, we do need to make sure we're consistent with how we think about 'closeness', in the same way we needed to make sure our idea of distance was consistent for metric spaces.

  - Recall back to the first axiom of metric spaces. We say that there's some distance between every pair of points, except when the points are actually the same point, in which there's no distance between them. We can translate this idea in terms of 'closeness' as requiring that the empty set $\varnothing$, and the entire set X must be elements of the topology $\mathcal{T}$ i.e. if we don't pick a point then there can't be any points close to it, and given two points exist in a set then on some scale they can be considered close, like I'm pretty close to shaking hands with Ryan Gosling given the scale of the entire universe. (The empty set is simply the set that contains no elements).
  - Let's say we have two regions of closeness in our topology, then if we combine these regions and consider it to be one region, then this should still be regarded as some region of closeness still. That is given any possible number of neighbourhoods, combining them all should also be considered as a neighbourhood; we can glue together collections of nearby points. Mathematically, given two open sets $M, N \in \mathcal{T}$, then $M \cup N \in \mathcal{T}$ as well.
  - In a similar vein, if a point lies in two different neighbourhoods at the same time, then it makes sense that it's also close to all of the points in the overlapping section of the two neighbourhoods; it's close in both senses simultaneously. Again mathematically, given two open sets $M, N \in \mathcal{T}$, then $M \cap N \in \mathcal{T}$ as well. NOTE: a very subtle point is that overlapping closeness only makes sense for a *finite* number of open sets. This is because if we could do so, we could shrink these open sets so much that we'd lose enough 'wiggle room' in the end product to regard any sort of closeness. As an example, consider $X = (-1, 1)$ the open interval between -1 and 1, and let $\mathcal{T}$ be all of the intervals that contain 0. There are an infinite number of these intervals, and taking all of their intersections leaves us just with the set containing 0, $\{0\}$. But this singleton set doesn't capture our idea of 'closeness', since it only contains a single point - there's no other points it's close to, so it's slightly preposterous to consider it as a region of close points. (Mathematically we can actually include singleton sets that is consistent with the axioms, like the discrete topology, but corresponds to a very different notion of 'closeness').

- As long as we select subsets of X that align with these axioms do we get a consistent idea of closeness in this space. A set of subsets satisfying these axioms is what we call a **topology** on X, and pairing the two together, $(X, \mathcal{T})$, we get a **topological space**. Notice that everything we've said about closeness depends only on which sets count as neighbourhoods, not on distances themselves. This suggests we can forget about distance entirely and just keep track of these sets.

- What's nice about how we built up topological spaces is that every metric space induces a topological space. We just let the open sets be all of the open balls in the ambient space as defined by the distance metric of that space. The most common one is aptly named the standard topology, which is defined over any Euclidean space $\mathbb{R}^n$ and the topology is taken to be all of the open balls in that space (and all of their unions and finite intersections). But, we can go beyond that, and define topological spaces which aren't induced by a metric space. (maybe give example?)

- Now with the idea of closeness in a space well-defined, we can finally define continuity without any knowledge of distance in a space. Like we said before, continuity is really just asking if there's a collection of points close to $f(a)$, then there's a collection of points around a that get mapped into the first collection. Mathematically: $f : X \to Y$ is continuous at $a \in X$ if for each neighbourhood $N \subseteq Y$ of $f(a)$, there's a neighbourhood $M \subseteq X$ such that $f(M) \subseteq N$. This is really cool! Now we have our notion of continuity purely described in terms of open sets(and their pre-images)- no metric needed!

- This definition is fine, but it's slightly cumbersome, as there are 2 qualifiers we'd need to find to verify a function is continuous (the M and N). Is there perhaps a natural neighbourhood M we can select instead? What if we took M to be all of the points that get mapped to N? This is known as the **pre-image** of N, $f^{-1}(N) = \{x \in X \mid f(x) \in N\}$. Now our definition says: $f : X \to Y$ is continuous at $a \in X$ if for each neighbourhood $N \subseteq Y$ of $f(a)$, $f^{-1}(N)$ is a neighbourhood of a. Now we only have one quantifier N!

- This is a really nice definition now. So to prove whether a function f is continuous everywhere, we just need to check that the pre-image of any open set in Y is also an open set. This is what we'd been building up to all this time - a way to define continuity while stripping away any idea of distance at all.

2.5 Homeomorphisms

- Now with a way to characterise continuous functions of topological spaces, how can we compare to spaces and say whether they are the same space topologically? Very simple actually! Say we want to determine whether spaces X and Y are the 'same', then we just need to find a bijective function $f : X \to Y$ such that f and its inverse f^{-1} are both continuous (just the function f being continuous is not enough to get continuity in the other direction automatically, my lecture notes has an example if needed). This is equivalent to requiring pre-images of open sets in Y under f are to open sets in X, and vice versa for f^{-1}. If there exists a homeomorphism between X and Y, then they're considered to be **homeomorphic** aka the same space.

- However in practise, it might be difficult to find an explicit homeomorphism between two spaces that are in fact homeomorphic, or directly prove that one doesn't exist. Luckily for us, there's lots of properties a topological space can have that don't change under a homeomorphic , these are called **topological invariants**. A very simple invariant is finiteness: finite spaces remain finite under homeomorphisms (because homeomorphisms are bijections). There are many such invariants such as compactness, connectedness, Hausdorff, metrisable. Invariants are super useful, as they allow us to rule when two spaces are **not** the same.

- So we've finished, we're able to determine whether two given spaces are the same under continuous transformations.

2.6 Not quite everything we need?

- There is one slight problem with homeomorphisms however, in that they don't quite conform to what we expect we should be able to do when deforming spaces. For instance, we can clearly take a cylinder (without the top and bottom) and compress it into a circle, and vice

versa (stretch the circle into a cylinder). However, these two spaces are **not** homeomorphic. Why? The simplest explanation I like is to observe what happens when you remove points from these spaces. The circle and cylinder are *connected* spaces, meaning if we pick any two points on them, we can connect them by drawing a continuous line between the two (this isn't exactly the definition of connectedness, but in this specific example this is an equivalent definition). Now let's remove two points from each of the spaces. The circle has now become disconnected, turned into two separate segments that we cannot draw a path between. However, on the cylinder we can still very easily draw paths between any two points on the cylinder. This means the two spaces cannot be homeomorphic.

- In fact, the circle and cylinder are **homotopy equivalent** to each other. Homotopy equivalence is a weaker version of being homeomorphic, which involves finding a continuous **homotopy** between two spaces, and allows for squishing and stretching of spaces, like our circle-cylinder example. There's lots more to talk about with homotopy equivalence, but let's save that for another video.